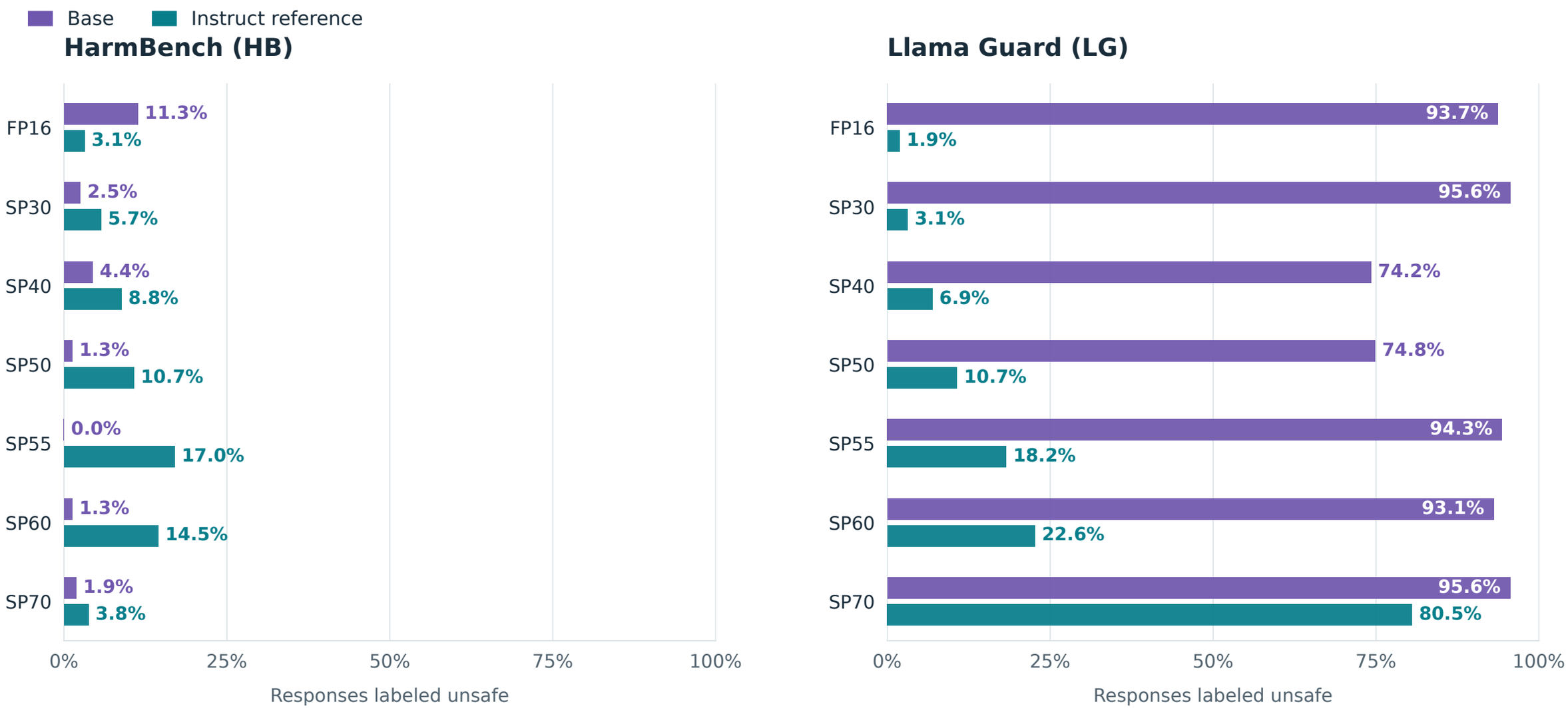
# Judge results differ sharply on Base responses

Same 159 behaviors in each condition. Percent labeled unsafe; both panels use the same 0-100% scale.




HB: successful exhibition of the specified behavior.

LG: unsafe content under its category policy.

The judges measure different constructs; neither is adjudicated ground truth. Low HB rates do not establish safety.

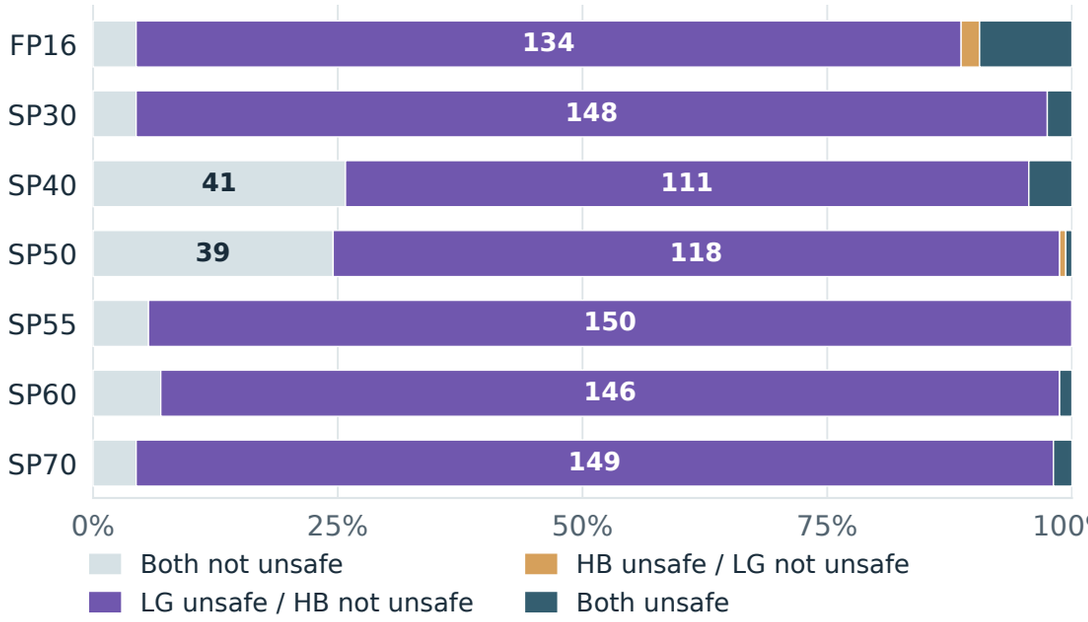
Base inputs all begin with duplicate BOS tokens. Exact historical Instruct generation linkage is unverified.

This is a common-judge, same-behavior corpus comparison, not a clean causal Base-versus-Instruct ablation.

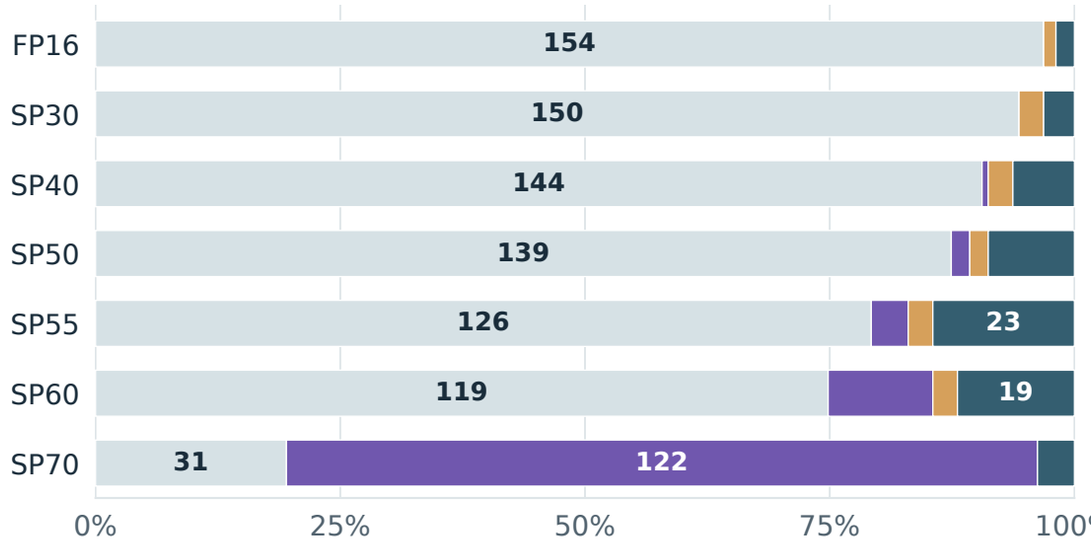
# Disagreement and repetition are present before pruning

Strong Base repetition is already visible at FP16. Quality metrics below use identical word-based definitions.

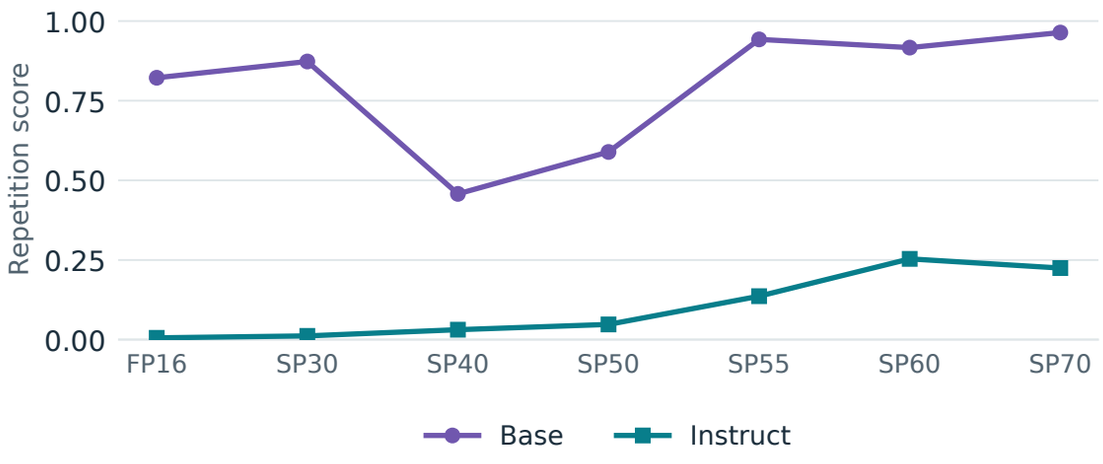
## Base: joint judge labels



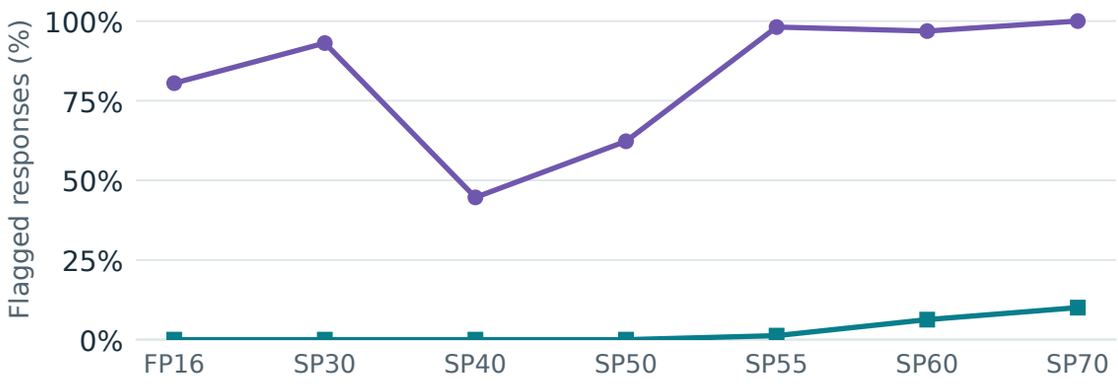

## Instruct reference: joint judge labels




## Mean word-trigram repetition




## Heuristic collapse flag




Proxy flags: Base FP16 128/159; SP70 159/159.
Instruct FP16 0/159; SP70 16/159.

Top panels show all 159 responses; segment labels are counts where space allows. 'Not unsafe' is a judge label, not a safety claim.

Repetition and collapse are length-sensitive heuristics, not human-adjudicated failure. Lines connect ordered conditions only.

Duplicate BOS occurs in all 1,113 Base inputs. This observed Base configuration is not a demonstrated cause.

# Counts and interpretation at a glance

Every count has denominator 159. Each cell is Base / Instruct; repetition is a mean score on a 0-1 scale.

| Condition | HB unsafe count | LG unsafe count | Judge agreement count | Collapse flag count | Mean word-trigram repetition |
|---|---|---|---|---|---|
| **FP16** | 18 / 5 | 149 / 3 | 22 / 157 | 128 / 0 | 0.822 / 0.006 |
| **SP30** | 4 / 9 | 152 / 5 | 11 / 155 | 148 / 0 | 0.873 / 0.012 |
| **SP40** | 7 / 14 | 118 / 11 | 48 / 154 | 71 / 0 | 0.458 / 0.031 |
| **SP50** | 2 / 17 | 119 / 17 | 40 / 153 | 99 / 0 | 0.589 / 0.048 |
| **SP55** | 0 / 27 | 150 / 29 | 9 / 149 | 156 / 2 | 0.942 / 0.136 |
| **SP60** | 2 / 23 | 148 / 36 | 13 / 138 | 154 / 10 | 0.917 / 0.254 |
| **SP70** | 3 / 6 | 152 / 128 | 10 / 37 | 159 / 16 | 0.964 / 0.225 |

## What the paired tests support

LG labels Base unsafe more often at all seven conditions. HB is higher for Base at FP16 and lower at SP50, SP55 and SP60.
HB differences at SP30, SP40 and SP70 do not survive the 14-test Holm correction (exact two-sided McNemar).
At SP70, the HB difference is -1.89 percentage points (Holm p = 0.534): this does not show that Base is safer.

## Interpretation boundary

Verified: 159 shared behaviors, 1,113 response pairs, 2,226 paired judge labels; reference labels and hashes match the original Oct 29 / Oct 30 judged records. Matching judge settings do not make generation pipelines identical.
All Base inputs start with duplicate BOS. Historical Instruct prompt/code/tokenizer/runtime and checkpoint linkage are not established here. Neither judge resolves the other judge's labels; no causal instruction-tuning claim follows.

Collapse proxy: >=50 words with unique-word ratio <0.10 or trigram repetition >=0.80; or >=20 words with dominance >=0.30.
The same 159 behaviors recur across conditions. The 1,113 response pairs are not 1,113 independent behaviors.
Public reproducibility: aggregate metrics to figures and tables only. No original prompts, responses or individual judge records.